# DATA ITEM DESCRIPTION

**Title**: NUCLEAR SURVIVABILITY TEST PLAN

**Number**: DI-NUOR-80928A

**Approval Date**: 20130201

**AMSC Number**: F9322

**Limitation**:

**DTIC Applicable**: N/A

**GIDEP Applicable**:

**Preparing Activity**: 27 (AFNWC)

**Applicable Forms**: N/A

**Use/Relationship**: The Nuclear Survivability Test Plan describes the detailed procedures governing the planning, preparation, implementation, and reporting of nuclear weapons effects simulation testing that will constitute adequate proof, when combined with data analysis, that an acceptable level of nuclear survivability has been achieved.

a. Application. When a solicitation or contract contains a requirement to demonstrate compliance with nuclear survivability requirements, this Data Item Description (DID) shall be listed on the Contract Data Requirements List (CDRL) (DD Form 1423). The documents sited in this DID are to be used for guidance and are not mandatory for the CDRLs.

b. This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirements as delineated in the contract.

c. When a solicitation or contract has this test plan as a requirement, DI-NUOR-80929, *Nuclear Survivability Test Report* shall also be listed on the DD Form 1423.

(Copies of the DID are available online at https://assist.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Ave., Bldg 4D, Philadelphia PA 19111-5094.)

d. Air Force Instruction (AFI) 10-2607, *Air Force Chemical, Biological, Radiological and Nuclear Survivability*, contains the AF policy applicable to nuclear survivability.

(Copies of the AF Instruction are available online at http://www.e-publishing.af.mil/.)

**Requirements**:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices and revisions, shall be as specified in the solicitation or contract.

2. General. The Nuclear Survivability Test Plan shall provide the detailed User and Technical Test methods, to include test and analysis, to demonstrate that the system meets or exceeds its nuclear survivability specifications. The test plan shall be prepared consistent with the information contained in AFI 10-2607. The test plan shall include the following information and shall be used for piece part, circuit, subsystem, or system tests.

3. Format. The Nuclear Survivability Test Plan shall be in the contractor's format.

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4. Background.

a. A statement of how these tests support system validation and/or acceptance, and the need for each test, considering existing data and the system's survivability requirements shall be provided.

b. A discussion of critical issues to include trade-off alternatives and risk associated with the nuclear survivability tests, identifying critical component/circuit/system performance parameters and functions which will be assessed during tests shall be presented.

c. A discussion of what considerations will be made for the interactions of the Government furnished equipment (GFE) with the contractor furnished equipment (CFE) and the consequent impact of these considerations on this test plan.

5. Content.

5.1 Specifics described in the plan.

a. A listing of piece parts, materials, components, circuits, structures, sub-assemblies, assemblies, and systems to be tested with detailed information concerning tests and parameters to be measured before, during, and after the nuclear environment tests (an example test data sheet must be included in this plan);monitor points and rationale for choices; instrumentation for sample response measurements to be employed; ambient conditions; modes of test sample functioning during the test with rationale; number of test samples and controls; the physical orientations of the samples to the test environment (for system level tests, this includes GFE and CFE orientations); the number and kinds of spare parts and GFE required (including simulation facilities); dosimetry and calibration; and the data to be recorded. The plan for the test data analysis must be detailed. Particular emphasis shall be given to hardness critical items (HCI) and to mission essential equipment and shall include non-developmental and proprietary items. Remote operation and monitoring of equipment shall be shown in detail.

b. Justification and rationale for the simulation/test equipment selected shall be provided. A description of the test facilities to be employed to include special instrumentation and simulator/tests facility characteristics (to include for ionizing radiation environments: radiation type; beam energy spectrum and pulse width), environment measurement techniques, and calibration procedures. This shall include how the chosen simulator environments resemble and differ from specified environments, and how these differences are going to be dealt with regarding interpretation of the test results (and extrapolation to the specified level, where required).

c. A statement of the specific test levels and test sequences each with an assigned priority (as pertains to the collection of the most important test data), to be employed in each environment test, and the time between the irradiation and the parameter measurements. Assigning priorities shall be in recognition of the fact that unexpected problems often arise during testing, and scheduling difficulties result. The critical test issues shall therefore be given top priority in the test sequence.

5.2 Data analysis descried in the plan.

a. The plan shall define the criterion for acceptable data quality.

b. The plan shall describe the impact of the test data, to include operational impact, error budget, and allowable downtime.



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c. The plan shall define the methodology for interpolation or extrapolation of test exposures or environments to a calculation or analysis of how the system responds to the specification threat criteria.

6. Miscellaneous.

a. The sequence of the simulated nuclear environments tests and the specifics of the test schedule (months after award of contract, duration of tests, and transportation time between facilities).

b. The job categories and the number of people required for each test.

c. Other information necessary to adequately delineate the planned Nuclear Survivability Test.

7. End of DI-NUOR-80928A.



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